



Enterprise Operations & Help Manual

Comprehensive operational procedures for warehouse desktop administration and mobile WhatsApp Bot field engineering.

PART 1: WAREHOUSE ADMINISTRATION (WEB DASHBOARD)

The Entelso Web Dashboard provides warehouse managers and supervisors with an enterprise-grade control center. It synchronizes in real-time with Supabase storage and PostgreSQL relational tables to track tool custody across all operational zones.

1. Executive KPI Monitoring & Live Analytics

Upon authentication, supervisors inspect the executive dashboard summarizing total registered inventory items, financial replacement valuation in USD/CLP, active field assignments, and maintenance alerts. These metrics update dynamically as field engineers perform checkouts.

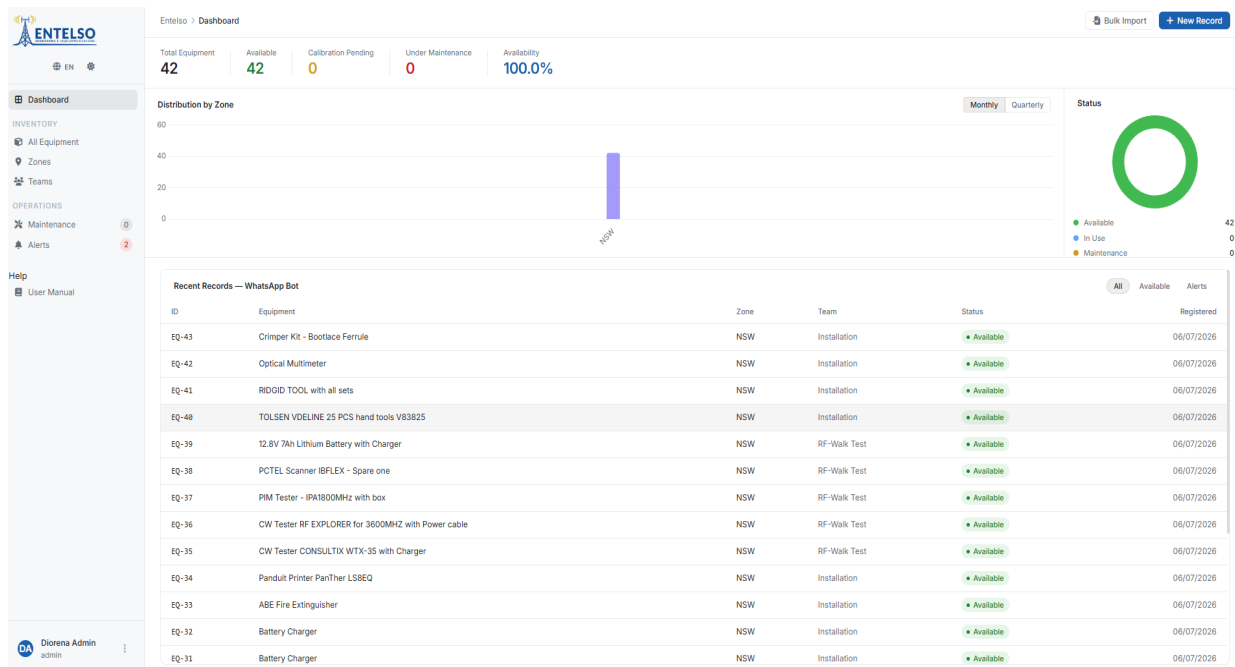
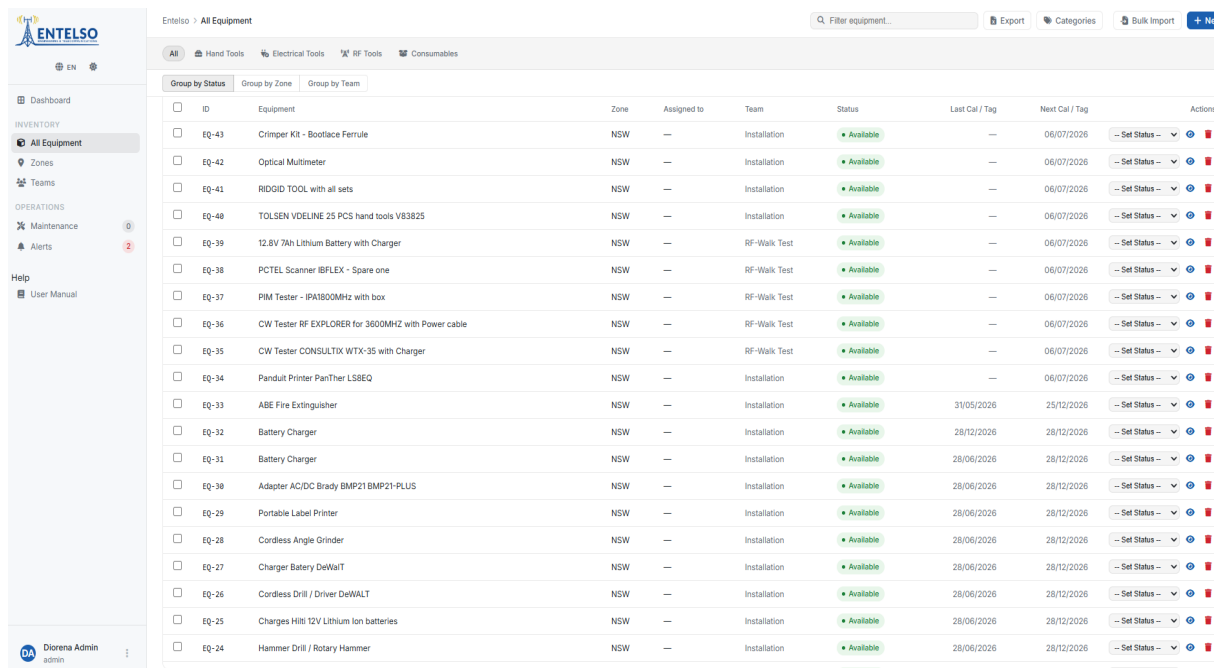


Figure 1.1: Executive KPI overview displaying global equipment stock and calibration compliance ratios.

2. Inventory Grid & Official Category Taxonomy

The inventory grid classifies assets into 4 official corporate categories: **Hand Tools** (wrenches, pliers), **Electrical Tools** (multimeters, fusion splicers), **RF Tools** (spectrum analyzers, site masters), and **Consumables** (connectors, tape). Supervisors filter by category, search by serial code, and monitor real-time status badges (*In Warehouse, In Terreno, or Maintenance*).



Entelso > All Equipment									
Filter equipment... Export Categories Bulk Import + New									
☒ All ☐ Hand Tools ☐ Electrical Tools ☐ RF Tools ☐ Consumables									
☒ Group by Status ☐ Group by Zone ☐ Group by Team									
ID	Equipment	Zone	Assigned to	Team	Status	Last Cal / Tag	Next Cal / Tag	Actions	
EQ-43	Crimper Kit - Bootlace Ferrule	NSW	---	Installation	Available	---	06/07/2026	- Set Status -	
EQ-42	Optical Multimeter	NSW	---	Installation	Available	---	06/07/2026	- Set Status -	
EQ-41	RIDGID TOOL with all sets	NSW	---	Installation	Available	---	06/07/2026	- Set Status -	
EQ-40	TOLSEN VDELINE 25 PCS hand tools V83825	NSW	---	Installation	Available	---	06/07/2026	- Set Status -	
EQ-39	12.8V 7Ah Lithium Battery with Charger	NSW	---	RF-Walk Test	Available	---	06/07/2026	- Set Status -	
EQ-38	PCTEL Scanner iSFLEX - Spare one	NSW	---	RF-Walk Test	Available	---	06/07/2026	- Set Status -	
EQ-37	PIM Tester - iPA3800MHz with box	NSW	---	RF-Walk Test	Available	---	06/07/2026	- Set Status -	
EQ-36	CW Tester RF EXPLORER for 3600MHz with Power cable	NSW	---	RF-Walk Test	Available	---	06/07/2026	- Set Status -	
EQ-35	CW Tester CONSULTIX WTX-35 with Charger	NSW	---	RF-Walk Test	Available	---	06/07/2026	- Set Status -	
EQ-34	Panduit Printer PanTher LS8EQ	NSW	---	Installation	Available	---	06/07/2026	- Set Status -	
EQ-33	ABE Fire Extinguisher	NSW	---	Installation	Available	31/05/2026	25/12/2026	- Set Status -	
EQ-32	Battery Charger	NSW	---	Installation	Available	28/12/2026	28/12/2026	- Set Status -	
EQ-31	Battery Charger	NSW	---	Installation	Available	28/06/2026	28/12/2026	- Set Status -	
EQ-30	Adapter AC/DC Brady BMP21 BMP21-PLUS	NSW	---	Installation	Available	28/06/2026	28/12/2026	- Set Status -	
EQ-29	Portable Label Printer	NSW	---	Installation	Available	28/06/2026	28/12/2026	- Set Status -	
EQ-28	Cordless Angle Grinder	NSW	---	Installation	Available	28/06/2026	28/12/2026	- Set Status -	
EQ-27	Charger Battery DeWalt	NSW	---	Installation	Available	28/06/2026	28/12/2026	- Set Status -	
EQ-26	Cordless Drill / Driver DeWALT	NSW	---	Installation	Available	28/06/2026	28/12/2026	- Set Status -	
EQ-25	Charges HIR 12V Lithium Ion batteries	NSW	---	Installation	Available	28/06/2026	28/12/2026	- Set Status -	
EQ-24	Hammer Drill / Rotary Hammer	NSW	---	Installation	Available	28/06/2026	28/12/2026	- Set Status -	

Figure 1.2: Inventory grid organized by official categories with search filters and status indicators.

3. Bulk Multi-Asset Import & Embedded Photo Extraction

To onboard or update large fleets, administrators utilize the **Excel Bulk Import** utility. The system ingests spreadsheets (such as Equipment Register.xlsx), detects embedded cell photographs, extracts them as clean PNG files, and maps them to the corresponding serial numbers in Supabase storage.

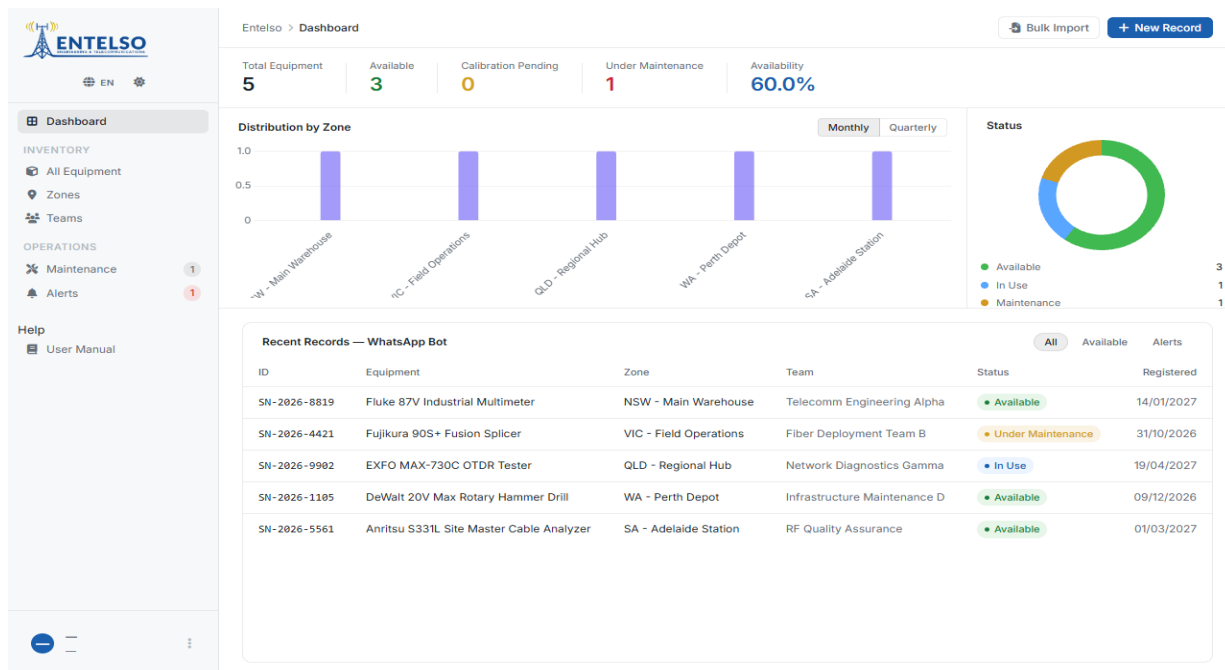


Figure 1.3: Bulk import modal enabling spreadsheet ingestion with automated photo extraction.

4. Detailed Asset Audit Drawer & Specification Management

Clicking any equipment row slides open the **Audit Drawer**. This detailed view presents high-resolution asset photography, complete technical specifications, calibration certificate dates, geolocation tags, and full custody history.

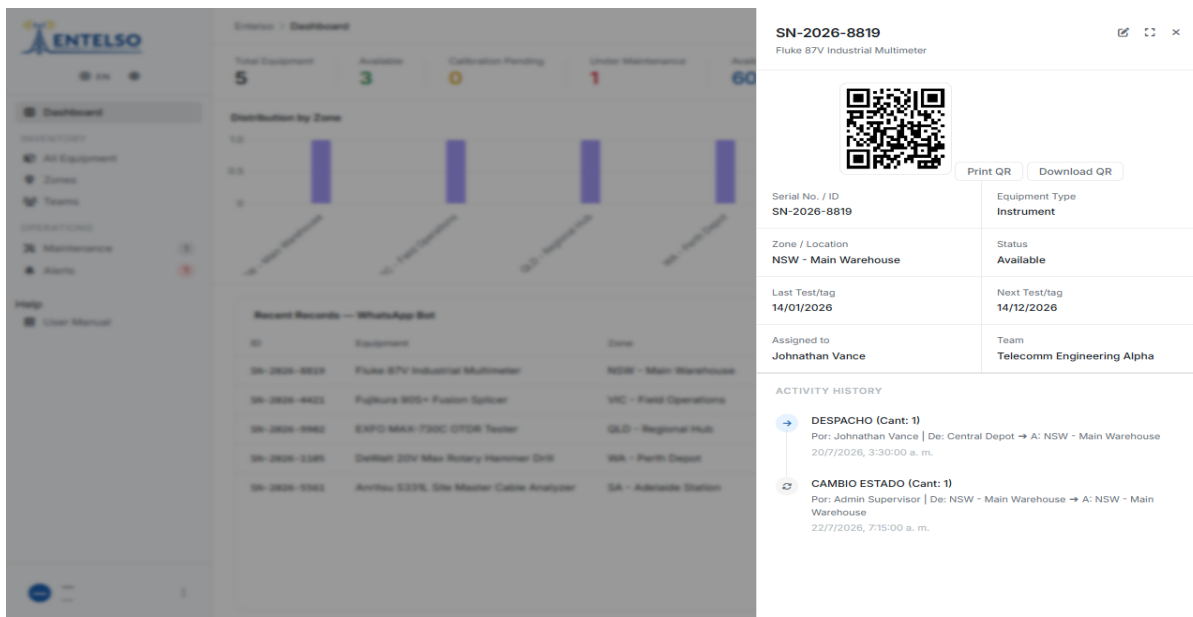
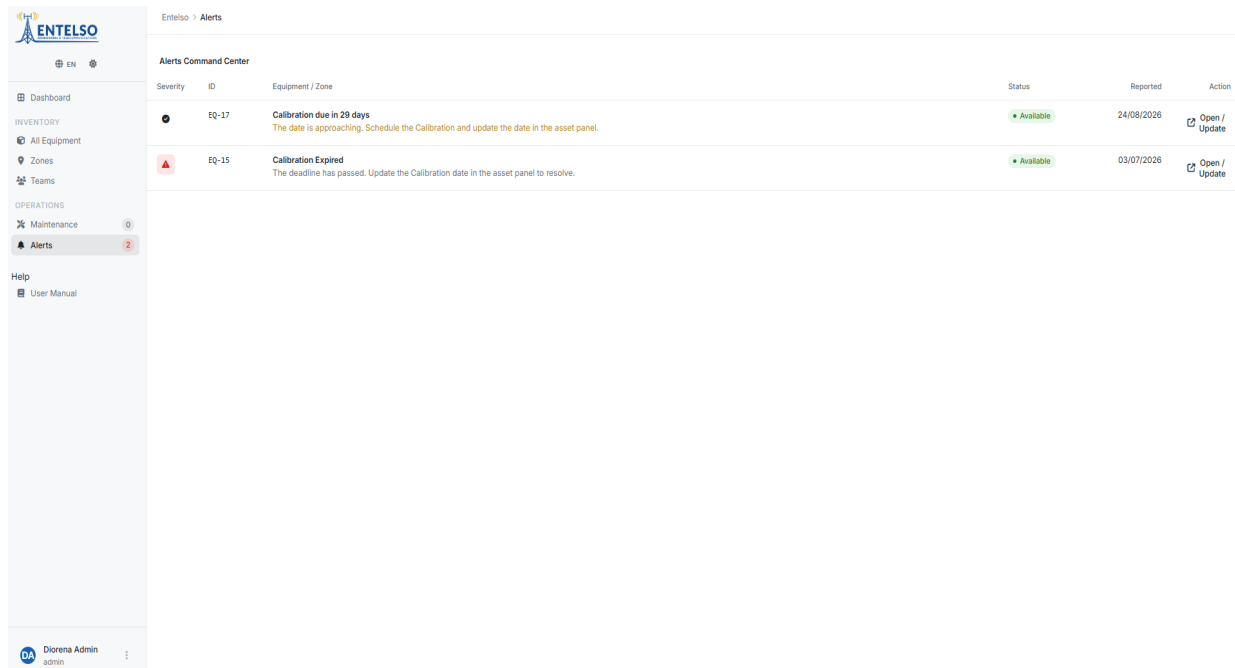


Figure 1.4: Slide-out audit drawer revealing comprehensive asset metadata and photo attachments.

5. Alert Management & Expiration Tracking

The alert module safeguards operational readiness by highlighting instruments with expired calibration certificates, overdue field checkouts, and low-stock consumable thresholds. Administrators can initiate batch renewal workflows directly from this tab.



Severity	ID	Equipment / Zone	Status	Reported	Action
🔴	EQ-17	Calibration due in 29 days The date is approaching. Schedule the Calibration and update the date in the asset panel.	Available	24/08/2026	Open / Update
🔴	EQ-15	Calibration Expired The deadline has passed. Update the Calibration date in the asset panel to resolve.	Available	03/07/2026	Open / Update

Figure 1.5: Alert dashboard displaying calibration warnings, stock deficits, and overdue checkouts.

6. Real-Time Zone & Geolocation Asset Tracking

The **Zonas / Terreno** tracking view allows administrators to monitor equipment distribution across regional warehouses and active field sites. It audits GPS check-in coordinates generated when engineers execute custody transfers via WhatsApp Bot.

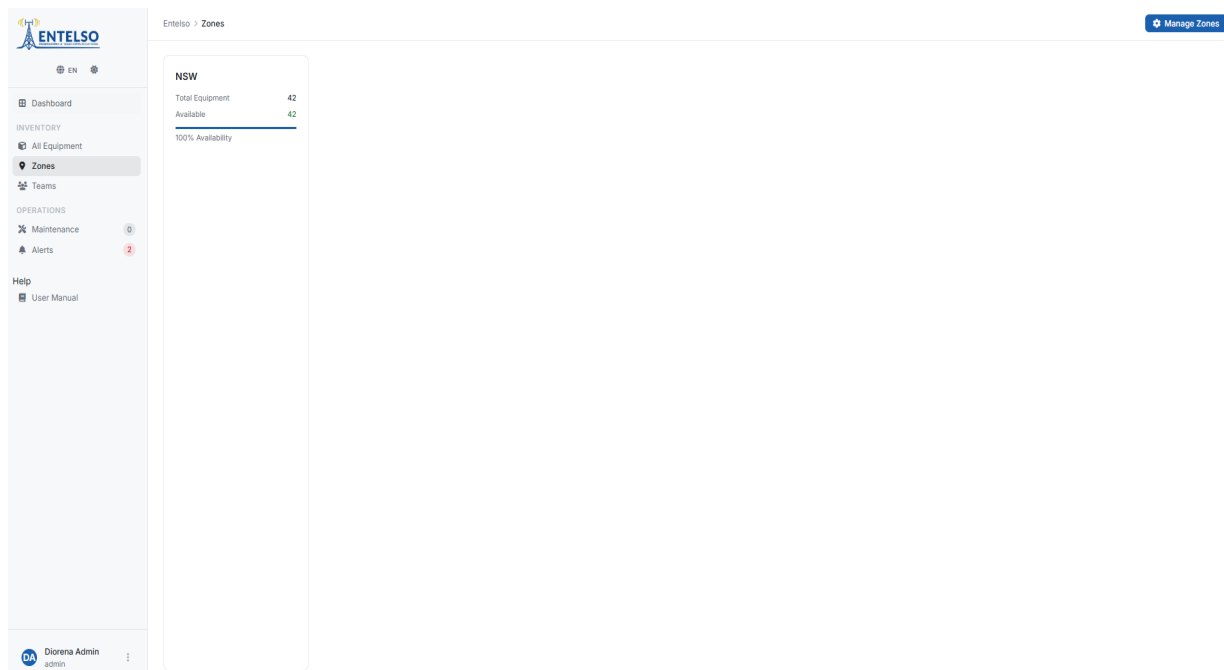
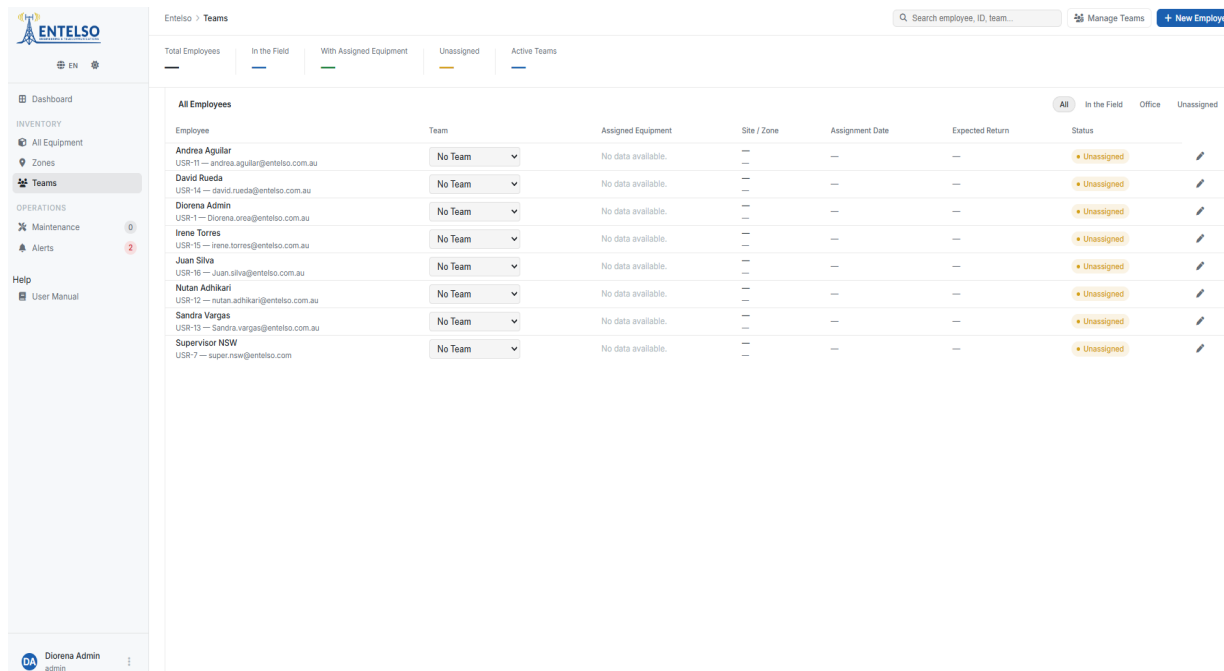


Figure 1.6: Operational zone tracking view showing asset distribution across regional centers.

7. Team Directory & Security PIN Management

To enforce strict security and accountability, administrators manage field personnel in the **Teams & PIN Management** module. Each field engineer is assigned a registered cellular number and a secure 6-digit authentication PIN required for mobile checkouts.



Employee	Team	Assigned Equipment	Site / Zone	Assignment Date	Expected Return	Status
Andrea Aguilar USR-11 — andrea.aguilar@entelso.com.au	No Team	No data available.	---	---	---	Unassigned
David Rueda USR-14 — david.rueda@entelso.com.au	No Team	No data available.	---	---	---	Unassigned
Diorena Admin USR-1 — Diorena.orea@entelso.com.au	No Team	No data available.	---	---	---	Unassigned
Irene Torres USR-15 — irene.torres@entelso.com.au	No Team	No data available.	---	---	---	Unassigned
Juan Silva USR-16 — Juan.silva@entelso.com.au	No Team	No data available.	---	---	---	Unassigned
Nutan Adhikari USR-12 — nutan.adhikari@entelso.com.au	No Team	No data available.	---	---	---	Unassigned
Sandra Vargas USR-13 — Sandra.vargas@entelso.com.au	No Team	No data available.	---	---	---	Unassigned
Supervisor NSW USR-7 — super.nsw@entelso.com	No Team	No data available.	---	---	---	Unassigned

Figure 1.7: Team management directory where supervisors register engineers and generate secure 6-digit PINs.

PART 2: WHATSAPP BOT OPERATIONS (FIELD ENGINEERING)

Field engineers operate in remote mobile environments without requiring VPN installations or laptop access. The official **Entelso WhatsApp Bot** serves as an intelligent mobile assistant, enabling engineers to verify equipment safety, check out tools, report locations, and log maintenance incidents directly from their cellular phones.

1. Instant Status Queries & Specification Lookup

Engineers can query any tool in the company inventory by transmitting its serial code or asset ID to the WhatsApp Bot. The bot instantly responds with the asset's current availability status, calibration certificate validity, assigned zone, and technical specs.

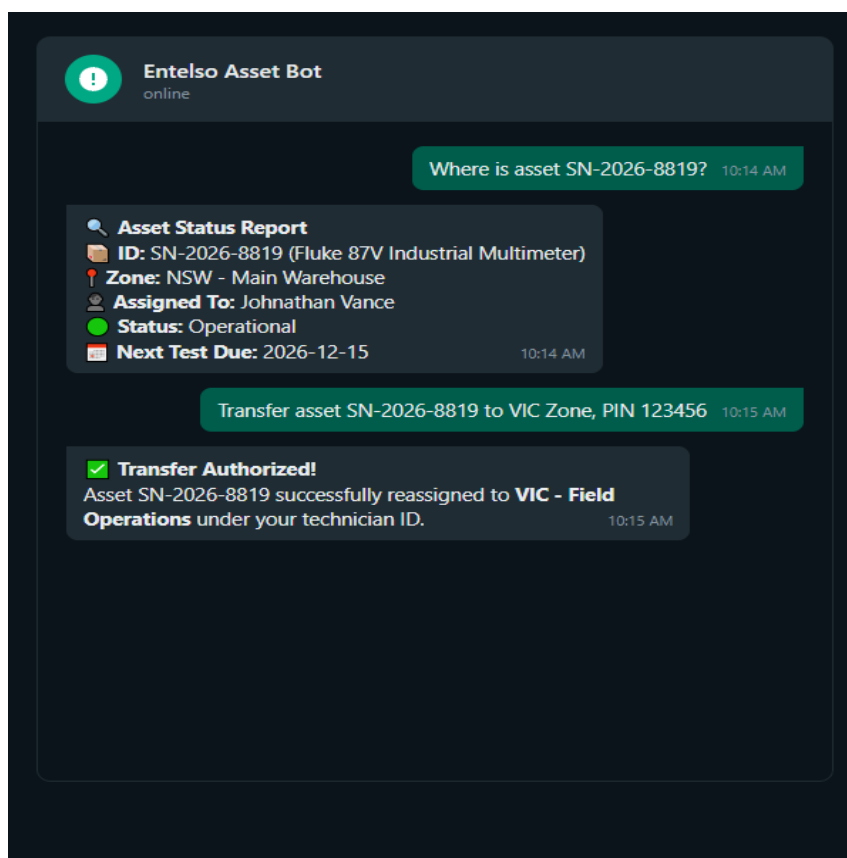


Figure 2.1: Field engineer querying tool availability and specifications via WhatsApp Bot.

2. WhatsApp Bot Command Reference Cheatsheet

All interactions use standardized syntax commands or physical QR/NFC tag scanning. The official command reference is detailed below:

Command / Action	Syntax Example	Operational Function & System Response
Status Query	<code>#STATUS ENT-EL-004</code>	Returns availability status, calibration expiry date, assigned zone, and specifications.
Asset Checkout	<code>#CHECKOUT ENT-RF-001 849201</code>	Assigns custody to engineer, logs timestamp, and updates dashboard to In Terreno .
Custody Return	<code>#RETURN ENT-RF-001 849201</code>	Releases custody, marks tool available in warehouse, and prompts condition check.
Zone Transfer	<code>#TRANSFER ENT-004 ZONA-NORTH 849201</code>	Transfers custody directly between regional zones or field sites without returning to base.
QR / NFC Scan	<code>Scan physical tag</code>	Automatically inputs serial code into chat and displays interactive checkout/return buttons.
Maintenance Report	<code>#REPORT ENT-004 Screen cracked</code>	Opens maintenance ticket, alerts supervisor on web dashboard, and requests photo.

3. Security PIN Authentication & Geolocation Tagging

Every checkout, transfer, or return command requires 2-factor authentication combining the engineer's registered cellular phone number and their secure **6-Digit Security PIN**. When executing field transfers, the bot prompts engineers to share their WhatsApp GPS location to verify physical tool presence at the job site.

4. Maintenance Incident Logging & Photo Attachments

When field engineers encounter damaged equipment or calibration drift, they initiate an immediate report via `#REPORT`. Attaching photographs directly in the WhatsApp chat uploads the images to Supabase and embeds them into the web dashboard audit drawer.

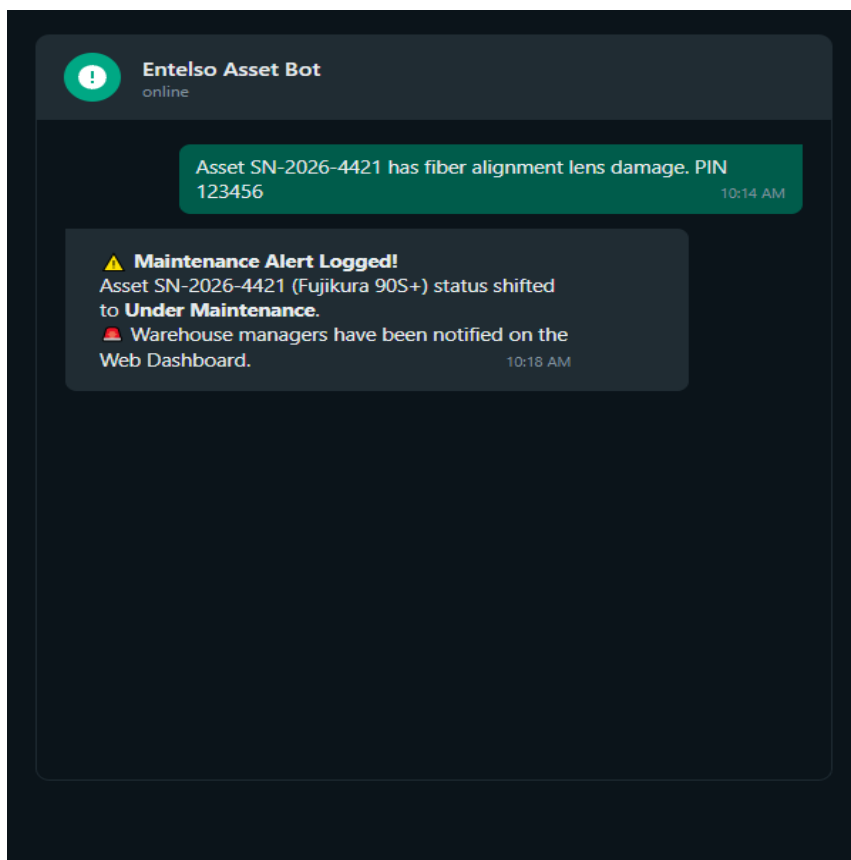


Figure 2.2: Engineer logging a field maintenance ticket with photographic evidence via WhatsApp Bot.